Astrikos AI Benchmark Report — Baseline Server vs Wildcat Lake

1. Summary

Comparison of the baseline server and Wildcat Lake. Each KPI below is the average of 5 runs per machine; the workload and method are described in section 2.

KPI	Baseline server	Wildcat Lake	Result
Inference throughput (samples/s, higher better)	515.2	707.4	+37.3%
Inference latency, mean (ms, lower better)	4.045	2.888	28.6% lower
Training throughput (samples/s, higher better)	134.8	240.0	+78.0%
Composite score S (higher better)	11.2862	15.6520	—

FINAL DELTA = +38.68% — Wildcat Lake scores ~1.39× the baseline on this workload.

```
S_baseline = sqrt(515.2007 / 4.0447) = 11.2862
S_wildcat  = sqrt(707.4333 / 2.8877) = 15.6520
Delta %    = (15.6520 / 11.2862 - 1) * 100 = +38.68%
```

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2. What the benchmark does

The workload is a subset of the Astrikos CPU-heavy MLflow pipelines (training + inference). It runs fully offline and is pure CPU — no GPU is used. The same pipelines and parameters were run on both machines, and we keep the work as close to byte-identical as possible so the comparison is fair. Each of the 5 runs has three phases:

1. **Training** — one full pass over the data under load → training throughput (samples/s).
2. **Inference throughput (KPI 1)** — large batches under sustained load → samples/s, higher better.
3. **Inference latency (KPI 2)** — many single-sample inferences, timed after a warmup → per-request ms (mean, p50, p95), lower better.

The 5 runs are averaged. The two KPIs are then combined into one higher-is-better score, and the delta is the ratio of the two machines' scores:

```
S       = sqrt( Throughput / Latency_ms )
Delta % = (S_wildcat / S_baseline − 1) × 100
```

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3. System configuration

	Baseline server	**Wildcat Lake**
CPU (physical host)	Intel Xeon 6 Performance 6515P — 16C/32T, 2.3 GHz, 72 MB cache, 150 W, DDR5-6400	Intel Core 5 330 — 6 cores, up to 4.8 GHz
Cores / threads used	6 / 6	6 / 6
Total memory	16 GB	16 GB
GPU	none (CPU-only by design)	none (CPU-only by design)
Inference engine	identical CPU build on both	identical CPU build on both
OS / kernel	Linux 6.8.0-124-generic	Linux 6.17.0-23-generic
Run date	2026-06-30	2026-06-30

Note: the baseline ran in a VM with 6 vCPUs on the Intel Xeon 6 6515P host; Wildcat Lake ran on bare metal. Both used 6 cores / 6 threads.

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4. Per-run results

Baseline server — 5 runs

Run	**Train tput (s/s)**	**KPI1 Throughput (s/s)**	**KPI2 Latency mean (ms)**	**p50 (ms)**	**p95 (ms)**
1	135.2	496.3	3.751	3.404	5.500
2	136.0	507.9	3.948	3.338	5.860
3	131.2	584.4	4.763	3.367	9.064
4	135.7	481.4	3.872	3.363	5.540
5	135.6	506.0	3.889	3.383	5.761
Avg	**134.8**	**515.2**	**4.045**	—	—

Composite score S = **11.2862**.

Wildcat Lake — 5 runs

Run	**Train tput (s/s)**	**KPI1 Throughput (s/s)**	**KPI2 Latency mean (ms)**	**p50 (ms)**	**p95 (ms)**
1	266.9	711.2	2.947	1.765	7.959
2	221.8	711.5	2.867	1.759	7.232
3	241.0	710.8	3.217	2.075	8.291

Run	**Train tput (s/s)**	**KPI1 Throughput (s/s)**	**KPI2 Latency mean (ms)**	**p50 (ms)**	**p95 (ms)**
4	229.8	700.7	2.679	2.068	6.682
5	240.2	702.9	2.727	2.023	7.185
Avg	**240.0**	**707.4**	**2.888**	—	—

Composite score S = **15.6520**.